

Seyi R. Afolayan

📍 Itasca, IL 60143 ✉️ seyirafolayan@gmail.com ☎️ +1 (443) 310-5548 in SeyiR 🌐 SeyiRPortfolio

Robotician and Controls Engineer with expertise in **motion planning**, **robot perception**, **real-time control**, and **sim-to-real deployment** of machines & industrial robotic systems. Strong **C++17** foundation with production experience in **ros2_control**, **MoveIt 2**, hardware interface development, and **force/compliant control**. Background spans **manipulation (UR5/UR5e, collaborative robots)**, perception(DL), mobile robots, UAVs, and underwater vehicles, alongside **industrial automation, PLC programming** (Allen-Bradley, Studio 5000), and **I/O integration** in manufacturing environments. Proven ability to deploy algorithms from simulation to **validated production hardware**, with emphasis on reliability, safety, and deterministic real-time response.

Education

The Johns Hopkins University <i>Master of Science in Engineering, Robotics (Control & Perception Track)</i>	<i>Baltimore, MD, USA</i> <i>08/2023 – 05/2025</i>
University of Prince Edward Island <i>B.Sc.Eng., Sustainable Design Engineering (Mechatronics) — Summa Cum Laude</i>	<i>Charlottetown, PEI, Canada</i> <i>09/2019 – 04/2023</i>

Technical Skills

Languages	C/C++, Python, MATLAB, Structured Text (IEC 61131-3), Ladder Logic, FBD
Middleware	ROS2 (Humble), MoveIt 2, Nav2, ros2_control , actionlib
Motion Planning	OMPL (RRT, PRM, BiEST), trajectory generation, collision checking
Perception & Vision	OpenCV, PyTorch, COLMAP (SfM), 3D Reconstruction (NeRF, 3DGS), camera calibration, object detection, RGB-D sensing
Industrial Auto.	Allen-Bradley PLCs, Studio 5000 / RSLogix 5000, FactoryTalk View (ME/SE), AutoCAD Electrical, EtherNet/IP, Modbus, FAT/SAT
Simulation & Viz	Gazebo/Ignition, NVIDIA Isaac Sim, URDF/SDF, sim-to-real pipelines, RViz
Robot Platforms	UR5/UR5e, FANUC, TurtleBot4, UAVs, ROV/AUV
Dev Tools	Git, Docker, Linux, CMake, CI/CD

Work Experience

Controls Engineer — Original Equipment Manufacturer	<i>The RDI Group, Itasca, IL</i> <i>09/2025 – Present</i>
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- Design electrical power and control schematics for OEM industrial manufacturing systems
- Program Allen-Bradley PLCs (Studio 5000 / RSLogix 5000) and design I/O architectures for industrial automation systems across multi-vendor manufacturing environments
- Develop AI-powered engineering tool using computer vision and LLMs to automate wire data extraction from electrical schematics, reducing manual data entry by approximately 80%
- Design and validate safety-critical control logic and communication protocols (EtherNet/IP, Modbus) for manufacturing systems requiring deterministic real-time response
- Develop HMI applications in FactoryTalk View for operator interfaces on production automation systems

Robotics Software Engineer Intern — Motion Planning & Simulation	<i>Johns Hopkins University</i> <i>06/2024 – 09/2024</i>
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Supervisors: Dr. Simon Leonard, Dr. Jin Seob Kim

- Developed motion planning and control algorithms in C++ using MoveIt 2 and ros2_control with validated sim-to-real deployment on UR5 hardware
- Built mobile manipulation platform integrating Nav2 navigation stack with arm trajectory generation for coordinated task execution
- Implemented sensor integration pipelines enabling perception-aware motion planning across Isaac Sim, Gazebo, and physical robot environments

Research Engineer — Underwater Robotics	<i>Dynamical Systems & Control Lab, JHU</i> <i>12/2023 – 08/2025</i>
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Supervisor: Dr. Louis Whitcomb (IEEE Fellow)

- Developed ROS2-based autonomy software for the JHUROV2 deep-sea underwater vehicle, implementing navigation and state estimation modules within the vehicle's autonomy stack
- Integrated ROS2 communication infrastructure and hardware interface drivers enabling real-time sensor data processing for subsea navigation under dynamic constraints

Aviation Systems Technician & Licensed Remote Pilot	<i>Cargojet, Canada</i> <i>04/2016 – 07/2019</i>
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- Performed systematic troubleshooting and validation for aviation systems, applying structured fault diagnosis and safety-critical testing methodology
- Operated UAV systems with emphasis on flight path planning, risk assessment, and regulatory compliance

Selected Projects

Real-Time Cartesian Controller for ros2_control — Controller	<i>C++17 Open Source</i>
- Developed <code>ros2_control</code> plugin in C++17 achieving ±0.7mm positioning accuracy at 500Hz for sensor-driven Cartesian motion on UR5e collaborative robot - Implemented damped pseudo-inverse Jacobian with KDL for robust singularity handling during continuous trajectory execution - Designed modular hardware interface architecture enabling seamless switching between NVIDIA Isaac Sim, Gazebo, mock hardware, and physical UR5e without code changes - Built automated CI testing framework validating controller stability, trajectory tracking performance, and sim-to-real consistency across all platforms	
Sampling-Based Motion Planning Pipeline for UR5/UR5e	<i>MoveIt 2 OMPL</i>
- Implemented and benchmarked BiEST, RRT, and PRM planners achieving 95%+ success rate for collision-free trajectory generation in cluttered manipulation environments - Built complete planning-to-execution pipeline: MoveIt 2 planning → trajectory validation → <code>ros2_control</code> execution on physical UR5 hardware - Validated motion plans across RViz, Isaac Sim, Gazebo, and physical UR5/UR5e hardware with consistent, reproducible performance metrics	
Neural Scene Reconstruction: NeRF and 3D Gaussian Splatting	<i>PyTorch Nerfstudio</i>
- Built end-to-end 3D scene reconstruction pipeline using COLMAP (Structure-from-Motion) for camera pose estimation followed by training with Nerfacto and Splatfacto (3D Gaussian Splatting) on the Nerfstudio framework - Evaluated reconstruction quality (sharpness, spatial accuracy) and training efficiency across three temporal sampling densities (150–600 frames), with Nerfacto yielding superior scene fidelity and Splatfacto achieving 40× faster rendering for real-time applications - Characterized trade-offs between input frame density, COLMAP processing time (2–30 min), and neural rendering quality to guide simulation data collection strategies	
Autonomous UAV Control System — Aurelia X4 Heavy-Lift Quadrotor	<i>Team Lead ROS2</i>
- Led 4-person team to first successful flight after multiple prior cohorts failed; designed complete ROS2 control architecture with autonomous waypoint navigation and trajectory tracking - Implemented sim-to-hardware deployment pipeline with MAVLink integration and NVIDIA Jetson Orin Nano / CubePilot for onboard real-time trajectory generation - Obtained FAA Part 107 Remote Pilot Certification specifically to serve as Remote PIC for live flight operations	
Autonomous Mobile Robot Navigation System	<i>Nav2 Behavior Trees</i>
- Developed custom Nav2 behavior tree for TurtleBot4 with dynamic obstacle avoidance, SLAM-based localization, and EKF state estimation - Validated full navigation stack across simulation and physical hardware with automated performance benchmarking	
Teaching & Mentorship	
Graduate Teaching Assistant	<i>Johns Hopkins University</i>
<i>EN.601.463/663: Algorithms for Sensor-Based Robotics — Instructor: Prof. Simon Leonard</i>	<i>08/2024 – 05/2025</i>
- Mentored 82+ graduate students implementing motion planning algorithms, manipulation, control theory, and UR5/UR5e hardware validation - Designed and graded programming assignments using automated unit-test frameworks emphasizing correct algorithm implementation and real-hardware deployment	
Teaching Assistant	<i>Johns Hopkins University</i>
<i>EN.530.343: Design and Analysis of Dynamic Systems — Instructor: Prof. Louis Whitcomb</i>	<i>12/2024 – 05/2025</i>
Led weekly problem-solving sessions on classical and modern control theory, servo systems, state-space methods, and stability analysis	
Graduate Teaching Assistant	<i>Johns Hopkins University</i>
<i>EN.530.430: Robot Sensors and Actuators — Instructor: Prof. David Kraemer</i>	<i>08/2024 – 12/2024</i>
Supervised hands-on laboratory sessions covering sensor calibration, integration, embedded systems (Arduino), and systematic hardware testing methodology	
Honors & Certifications	
LCSR Distinguished Scholarship, Johns Hopkins University	<i>08/2023 – 05/2025</i>
Full tuition and health insurance fellowship awarded for research excellence in robotics	
Engineers PEI Graduating Student Award	<i>05/2023</i>
Top graduating student, Sustainable Design Engineering	
Technical Excellence Award, Faculty of Sustainable Design Engineering, UPEI	<i>04/2023</i>
Fung Excellence Award, Coleman Fung Institute, UC Berkeley	<i>03/2023</i>
Merit-based scholarship at maximum value (Declined)	
Mastercard Foundation Scholars Program, UC Berkeley — Finalist (Declined)	<i>03/2023</i>
Remote Pilot Certifications: FAA Part 107 & Transport Canada Unmanned Aircraft Pilot Certificate	<i>01/2025, 09/2020</i>